

Power BI Sales Analytics Dashboard

Project Overview

This project demonstrates the end-to-end development of a scalable Power BI data model and interactive sales analytics dashboard. The primary objective was to transform raw, fragmented data into a clean, relational star schema capable of supporting accurate reporting, advanced analytics, and secure access.

The source data consisted of multiple disconnected and inconsistently structured tables containing data quality issues that required extensive transformation before meaningful analysis could be performed.

Project Objectives

- Transform messy source data into a clean analytical model
- Design and implement a scalable star schema
- Improve data quality through validation and anomaly detection
- Build a secure reporting environment using Row-Level Security (RLS)
- Create a reusable date dimension for time intelligence
- Develop a model optimized for future expansion and dashboard reporting

Data Preparation

The original datasets contained inconsistencies, duplicate structures, and scattered information across multiple tables.

Data preparation included:

- Cleaning and standardizing raw datasets
- Renaming tables and columns using consistent naming conventions
- Merging and appending disparate datasets into unified fact tables
- Creating well-structured dimension tables
- Detecting and correcting data anomalies through exploratory data analysis
- Standardizing data types and formats

Data Modeling

A scalable relational data model was designed following dimensional modeling best practices.

Key modeling activities included:

- Designing a star schema
- Creating dimension and fact tables

- Generating surrogate/primary keys for dimension tables
- Importing corresponding foreign keys into fact tables
- Establishing one-to-many relationships between dimensions and facts
- Creating a centralized Date dimension shared across multiple fact tables
- Configuring relationship cardinality and filter direction to ensure correct aggregation and filtering
- Optimizing the model for scalability and future enhancements

One of the major learning outcomes during this project was understanding how relationship cardinality and cross-filter direction directly affect calculation accuracy in Power BI. Incorrect relationship configurations produced misleading aggregations, which were resolved by implementing proper one-to-many relationships with appropriate filter propagation.

Security Implementation

Implemented dynamic Row-Level Security (RLS) to restrict data visibility based on user identity.

Security features include:

- Dynamic filtering using the logged-in user's email address
- Region-based access control
- Dedicated Security table for authorization mapping
- Automatic filtering of fact tables through model relationships

Technical Skills Demonstrated

- Power BI
- Power Query (M)
- Dimensional Data Modeling
- Star Schema Design
- Fact & Dimension Modeling
- Primary and Foreign Key Design
- Relationship Management
- Data Cleaning
- Data Transformation
- Data Exploration
- Data Quality Validation

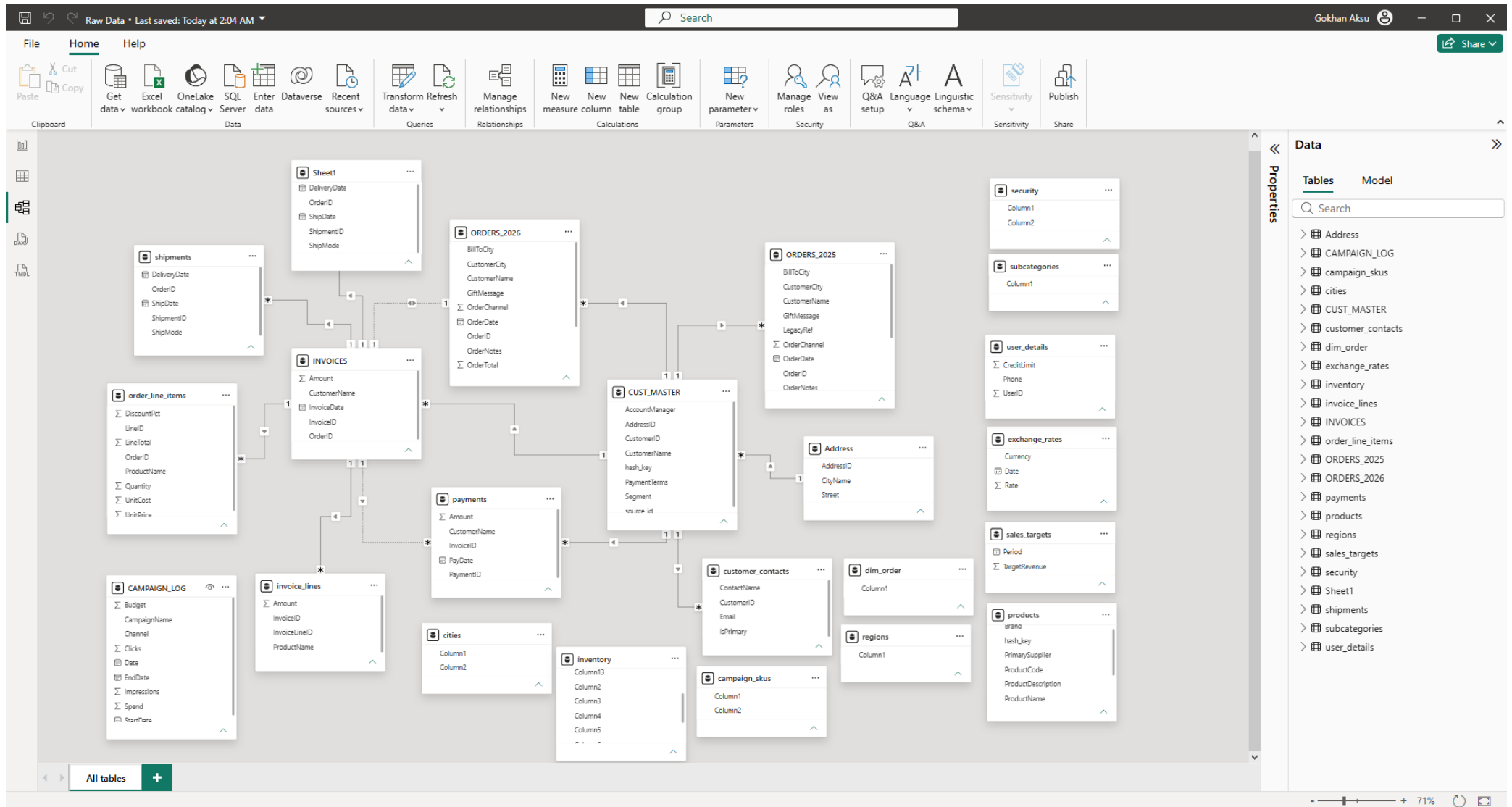
- Row-Level Security (RLS)
- Date Dimension Design
- Time Intelligence Preparation
- Dashboard Data Architecture

Key Learning Outcomes

This project strengthened practical skills in:

- Building production-ready Power BI data models
- Designing scalable star schemas
- Resolving relationship and cardinality issues
- Implementing secure, role-based reporting
- Structuring datasets for efficient analytics
- Applying dimensional modeling best practices used in enterprise Business Intelligence solutions

Raw Data Model



Star Schema Data Model

